Lab Report – EE-170L Computer Programming (Lab 14)

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LAB	14 cpp
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► Objectives

- To design a program that calculates GPA for multiple students.
- To implement a function that computes marks using 20% sessional, 30% mid, and 50% final.
- To calculate GPA relative to the class topper, ensuring fairness.
- To practice arrays, loops, and functions without using structures.

► Theory

In academic grading, GPA (Grade Point Average) is a weighted measure of student performance.

- Each subject has credit hours, which determine its importance in GPA calculation.
- Marks are divided into sessional (20%), mid (30%), and final (50%).
- GPA is calculated by converting marks into a 4.0 scale.
- In relative grading, the topper is assigned GPA = 4.0, and all other students are scaled proportionally:

Relative GPA=(student gpa /topper gpa) x4.0

This ensures the topper always gets the maximum GPA, and others are graded relative to them.

► Program Code

```
#include<iostream>
using namespace std;

// Function to calculate weighted marks for a subject
// Formula: 20% sessional + 30% mid + 50% final

double calculateMarks(double sessional, double mid, double final){
  return (sessional*0.20) + (mid*0.30) + (final*0.50);
}

// Function to convert marks into GPA (absolute scale)
// 100 marks = 4.0 GPA

double marksToGPA(double marks){
   double gpa = marks / 25.0; // divide by 25 to map 100 → 4.0
   if(gpa > 4.0) gpa = 4.0;  // cap GPA at 4.0
   return gpa;
}
```

```
int main(){

  int students;

   cout << "Enter number of students: ";
   cin >> students;


  string names[100]; //array to store student names
 double gpas[100];  //array to store each studentsGPA
 double topperGPA =0; // variable to track topper GPA

   // Loop for each student

  for(int s = 0; s < students; s++) {
     cout<<"\nEnter name of Student "<< s+1<< ": ";
      cin >> names[s];

     int subjects;
 cout<<"Enter number of subjects for"<<names[s]<<":";
      cin >> subjects;

 double totalPoints = 0, totalCredits = 0;
```

```
cout<<"\n— Enter Marks for"<< names[s]<<"—-\n";


   // Loop for each subject
for(int i = 0; i < subjects; i++) {
   double credit, sessional, mid, final;

      cout<<"\nSubject"<<i+1<<"credit hours: ";
   cin >> credit;

  cout<<"Enter sessional marks(out of 100): ";
   cin >> sessional;

 cout << "Enter mid marks (out of 100): ";
    cin >> mid;

 cout << "Enter final marks (out of 100): ";
   cin >> final;

 // Calculate subject marks using function
      double subjectMarks=calculateMarks(sessional,  mid,final);


    // Convert marks to GPA
double subjectGPA = marksToGPA(subjectMarks);


   // Show GPA for each subject
  cout<<"Subject"<<i+1<<"GPA ="<<subjectGPA<<endl;
```

```
// Add weighted GPA (GPA × credit hours)
totalPoints += subjectGPA * credit;
totalCredits += credit;


}

// Calculate overall GPA for student
gpas[s] = totalPoints / totalCredits;

// Update topper GPA if this student is higher
if(gpas[s] > topperGPA) topperGPA = gpas[s];


}

// Display relative GPA results
cout << "\n-- Relative GPA Results --\n";

for(int s = 0; s < students; s++) {

// Scale GPA relative to topper

double relative = (gpas[s] / topperGPA) * 4.0;

cout<<names[s]<<"Relative GPA="<<relative<<endl;

}

return 0;
}
```

► Conclusion

· The program successfully calculates absolute GPA for each student.

· It identifies the topper and scales all GPAs relative to them.

· This ensures fairness in grading, as performance is measured against the best student.

· The lab reinforced concepts of arrays, loops, functions, and relative grading logic.